MANAGEMENT OPERATING REVIEW - RECORD 003

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Quarterly Business Review for Q1 2024: purpose and attendance

Notes were captured live and circulated the same day, which ended a habit of arguing about what was decided. Pre-reads went out ahead of the 2024-04-25 session. Priya Ramanathan, VP Deployment, presented the operations section. Here the queue was read account by account. Only one hub held tooling then. The next session is set for 2024-05-16. One function submitted its pre-read late. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Guests from the deployment crews attended by video and contributed the only new information in the operations section. Notes were circulated on 2024-04-25. Priya Ramanathan, VP Deployment, presented the operations section. Here the queue was read account by account. Only one hub held tooling then. Pre-reads went out ahead of the 2024-04-25 session. The Quarterly Business Review convened for Q1 2024. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Reading the packet aloud was abandoned two sessions ago and nobody has asked for it back. The next session is set for 2024-05-16. The Quarterly Business Review convened for Q1 2024. Here the queue was read account by account. Only one hub held tooling then. Priya Ramanathan, VP Deployment, presented the operations section. Pre-reads went out ahead of the 2024-04-25 session. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Data definitions travel with the packet now, because two functions were counting the same thing differently. Notes were circulated on 2024-04-25. The Quarterly Business Review convened for Q1 2024. Here the queue was read account by account. Only one hub held tooling then. Tobias Nkemdirim attended as VP Engineering. Pre-reads went out ahead of the 2024-04-25 session. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Order book and bookings quality

Nobody in commercial believes the current forecast discipline is good enough, and the review agreed to inspect deal-level notes rather than roll-ups. The book took 142 units, against 38 last period. Signed volume: 142 units, win rate 26 percent. Here the queue was read account by account. Only one hub held tooling then. Intake closed at 142 units for Q1 2024. Open pursuits number 86, up from 79. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

The competitive picture shifted, with two losses attributed to a rival that had not appeared in a deal review for three periods. Win rate held at 26 percent on 86 live opportunities. Bexley Pharmaceutical Wholesale accounted for the largest single order. Here the queue was read account by account. Only one hub held tooling then. The book took 142 units, against 38 last period. Intake closed at 142 units for Q1 2024. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Commercial intake for the period came in ahead of the internal plan on unit count and behind it on gross margin, which is the trade the team was asked to make. Bexley Pharmaceutical Wholesale accounted for the largest single order. Open pursuits number 86, up from 79. Here the queue was read account by account. Only one hub held tooling then. Intake closed at 142 units for Q1 2024. The book took 142 units, against 38 last period. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Discounting concentrated in one competitive displacement rather than spreading across the book, which the review treated as an acceptable outcome. Win rate held at 26 percent on 86 live opportunities. Bexley Pharmaceutical Wholesale accounted for the largest single order. Here the queue was read account by account. Only one hub held tooling then. Intake closed at 142 units for Q1 2024. The book took 142 units, against 38 last period. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Late-quarter intake distorted the shape of the funnel again, and the pattern is now visible in three consecutive periods of data. Win rate held at 26 percent on 86 live opportunities. Discount depth peaked on the Alder & Vance Retail multi-site award. Here the queue was read account by account. Only one hub held tooling then. Open pursuits number 86, up from 79. Intake closed at 142 units for Q1 2024. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

A single reseller relationship produced more qualified volume than the whole outbound programme, which raises an uncomfortable question about channel mix. The book took 142 units, against 38 last period. Open pursuits number 86, up from 79. Here the queue was read account by account. Only one hub held tooling then. Win rate held at 26 percent on 86 live opportunities. Bexley Pharmaceutical Wholesale accounted for the largest single order. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Fleet uptime by live site, Q1 2024

Site	Fleet units	Uptime %	Contract floor %	Service credit exposure
Tulsa	14	96.6	96.0	none
Redlands	24	95.7	97.0	one week in credit
Grand Forks	16	96.2	97.0	credit claimed
Waukegan	18	95.5	96.0	one week in credit
Mesquite	20	96.3	97.0	credit claimed
Ashland Park	14	95.8	97.0	one week in credit
Riverbend	24	96.5	96.0	none
Kennesaw	22	96.0	97.0	one week in credit
Fort Dodge	18	96.7	97.0	credit claimed
Salinas Valley	16	95.6	96.0	one week in credit
Chandler South	22	96.4	97.0	credit claimed
Elk Grove	20	95.9	97.0	one week in credit
Cedar Bluff	14	96.6	96.0	none
Norfolk	24	95.7	97.0	one week in credit
Tulsa (phase two)	16	95.9	97.0	credit claimed
Redlands (phase two)	18	95.2	96.0	one week in credit
Grand Forks (phase two)	20	96.0	97.0	credit claimed
Waukegan (phase two)	14	95.5	97.0	one week in credit
Mesquite (phase two)	24	96.2	96.0	none
Ashland Park (phase two)	22	95.7	97.0	one week in credit

Deal inspection

The team agreed to stop reporting weighted pipeline in this meeting, on the grounds that nobody had ever acted on the number. Priya Ramanathan owns the re-qualification pass. Two losses went to the same competitor. Here the queue was read account by account. Only one hub held tooling then. Conversion into commercial close remains the weak stage. The Pinebank Distribution pursuit re-dated again. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Of the opportunities that left the quarter, most remain live and dated, and only a minority were lost to a competitor outright. Priya Ramanathan owns the re-qualification pass. Weighted coverage sits near 86 open pursuits. Here the queue was read account by account. Only one hub held tooling then. Deals leaving the period: 9, most still dated. The Pinebank Distribution pursuit re-dated again. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Two accounts that were treated as sure things at the last review have gone quiet, and neither seller has escalated. Weighted coverage sits near 86 open pursuits. The Pinebank Distribution pursuit re-dated again. Here the queue was read account by account. Only one hub held tooling then. Two losses went to the same competitor. Deals leaving the period: 9, most still dated. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Where a pilot converted, the common thread was an operations sponsor who owned a throughput target rather than a technology budget. Weighted coverage sits near 86 open pursuits. The Pinebank Distribution pursuit re-dated again. Here the queue was read account by account. Only one hub held tooling then. Conversion into commercial close remains the weak stage. Deals leaving the period: 9, most still dated. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Forecast accuracy at the regional level was worse than the aggregate, meaning two regions cancelled each other's errors out. The Pinebank Distribution pursuit re-dated again. Two losses went to the same competitor. Here the queue was read account by account. Only one hub held tooling then. Weighted coverage sits near 86 open pursuits. Priya Ramanathan owns the re-qualification pass. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

Stage conversion tells a cleaner story than the headline number does, because the drop is concentrated between technical validation and commercial close. Two losses went to the same competitor. The Pinebank Distribution pursuit re-dated again. Here the queue was read account by account. Only one hub held tooling then. Conversion into commercial close remains the weak stage. Slippage totalled 9 opportunities against 6 before. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Intervention causes across the fleet, Q1 2024

Cause category	Site band	Interventions	Share of total %	Accountable party	Trend
Floor marking and facility	ten largest sites	12	7.2	Manufacturing	flat
Operator intervention	ten largest sites	8	4.8	Field service	worsening
Navigation software	ten largest sites	6	3.6	Product	improving sharply
Drive unit hardware	ten largest sites	3	1.8	Customer facility team	first period measured
Sensor head hardware	ten largest sites	27	16.2	Customer operations	improving
Warehouse management interface	ten largest sites	19	11.4	Engineering	flat
Battery and charging	ten largest sites	16	9.6	Manufacturing	worsening
Unclassified	ten largest sites	12	7.2	Field service	improving sharply
Floor marking and facility	remaining sites	6	3.6	Product	first period measured
Operator intervention	remaining sites	4	2.4	Customer facility team	improving
Navigation software	remaining sites	3	1.8	Customer operations	flat
Drive unit hardware	remaining sites	2	1.2	Engineering	worsening
Sensor head hardware	remaining sites	15	9.0	Manufacturing	improving sharply
Warehouse management interface	remaining sites	11	6.6	Field service	first period measured
Battery and charging	remaining sites	9	5.4	Product	improving
Unclassified	remaining sites	7	4.2	Customer facility team	flat

Installation programme

Every site that slipped this period slipped for a reason on the customer side, which is a better position than last year and still not good enough. Cedar Bluff and Salinas Valley both cleared acceptance testing. Contract to go-live averaged 59 days. Here the queue was read account by account. Only one hub held tooling then. The queue carries 11 sites, 9 a period ago. Go-live cycle: 59 days, from 64. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand

in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Sequencing the queue by contractual date rather than by readiness produced two expensive standstills, and the rule has been reversed. Queue depth reached 11 sites. Cedar Bluff and Salinas Valley both cleared acceptance testing. Here the queue was read account by account. Only one hub held tooling then. Tobias Nkemdirim sequenced the queue by readiness. Contract to go-live averaged 59 days. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

A retrofit site behaved nothing like a greenfield site, and the estimate model still treats them as the same class of work. Cedar Bluff and Salinas Valley both cleared acceptance testing. Go-live cycle: 59 days, from 64. Here the queue was read account by account. Only one hub held tooling then. The queue carries 11 sites, 9 a period ago. Queue depth reached 11 sites. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Post-go-live stabilisation periods are running longer than the plan allows, so crews are being pulled back to sites already signed off. Norfolk was held at the readiness gate. Cedar Bluff and Salinas Valley both cleared acceptance testing. Here the queue was read account by account. Only one hub held tooling then. Contract to go-live averaged 59 days. Queue depth reached 11 sites. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Site queue depth grew again, and the queue is now long enough that sequencing decisions have commercial consequences rather than just scheduling ones. Contract to go-live averaged 59 days. Tobias Nkemdirim sequenced the queue by readiness. Here the queue was read account by account. Only one hub held tooling then. Cedar Bluff and Salinas Valley both cleared acceptance testing. Norfolk was held at the readiness gate. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

Readiness surveys are now a hard gate, and two sites were held at the gate rather than allowed to start with known network gaps. Contract to go-live averaged 59 days. Norfolk was held at the readiness gate. Here the queue was read account by account. Only one hub held tooling then. Cedar Bluff and Salinas Valley both cleared acceptance testing. The queue carries 11 sites, 9 a period ago. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Field reliability review

Uptime is measured differently by two customers than by us, and the review chose to adopt the stricter definition internally. Fleet uptime averaged 96.1 percent. Mean time between interventions was 281 hours. Here the queue was read account by account. Only one hub held tooling then. Renata Voss leads the joint remediation plan. Uptime reached 96.1 percent, from 96.4 percent. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Mean time between interventions is the number the service team manages to, and it moved in the right direction at every site but one. Bexley Pharmaceutical Wholesale measures uptime on a stricter basis. Cedar Bluff remains the weakest site on interventions. Here the queue was read account by account. Only one hub held tooling then. Uptime reached 96.1 percent, from 96.4 percent. Fleet uptime averaged 96.1 percent. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Facility conditions explain more of the intervention rate than any hardware factor, which took a year of data to establish convincingly. Renata Voss leads the joint remediation plan. Uptime reached 96.1 percent, from 96.4 percent. Here the queue was read account by account. Only one hub held tooling then. Interventions spaced 281 hours apart on the fleet median. Mean time between interventions was 281 hours. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Service credits remain the sharpest feedback loop the company has, and the review looked at exposure rather than at incurred credits alone. Renata Voss leads the joint remediation plan. Interventions spaced 281 hours apart on the fleet median. Here the queue was read account by account. Only one hub held tooling then. Bexley Pharmaceutical Wholesale measures uptime on a stricter basis. Mean time between interventions was 281 hours. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Intervention data is only as good as what the operators record, and two sites have stopped recording anything useful. Fleet uptime averaged 96.1 percent. Uptime reached 96.1 percent, from 96.4 percent. Here the queue was read account by account. Only one hub held tooling then. Cedar Bluff remains the weakest site on interventions. Interventions spaced 281 hours apart on the fleet median. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

A single navigation parameter set improved intervention rates at every site it was applied to, which is now the standard configuration. Fleet uptime averaged 96.1 percent. Uptime reached 96.1 percent, from 96.4 percent. Here the queue was read account by account. Only one hub held tooling then. Interventions spaced 281 hours apart on the fleet median. Bexley Pharmaceutical Wholesale measures uptime on a stricter basis. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Customer escalations live at the Q1 2024 review

Account	Issue	Opened	Severity	State	Next step
Pinebank Distribution	interface timeouts to the host	2024-04-13	high, customer executive involved	mitigated	parameter set applied
Alder & Vance Retail	sensor head replacement scheduling	2024-04-04	medium	closed	definition agreed
Northstar Cold Chain	operator training gap	2024-03-26	high, customer executive involved	monitoring	joint review scheduled
Havering Logistics	uptime reporting dispute	2024-03-17	medium	escalated to product	joint review scheduled
Quillon Home Supply	docking misalignment after release	2024-03-08	high, customer executive involved	open	parameter set applied
Tenmile Freight	aisle congestion at peak	2024-02-28	medium	mitigated	site visit booked
Bexley Pharmaceutical Wholesale	charger fault on two units	2024-02-19	high, customer executive involved	closed	parts shipped
Marrowfield Grocers	interface timeouts to the host	2024-02-10	medium	monitoring	site visit booked
Pinebank Distribution - second site	sensor head replacement scheduling	2024-02-01	high, customer executive involved	escalated to product	parameter set applied
Alder & Vance Retail - second site	operator training gap	2024-01-23	medium	open	site visit booked
Northstar Cold Chain - second site	uptime reporting dispute	2024-01-14	high, customer executive involved	mitigated	parts shipped
Havering Logistics - second site	docking misalignment after release	2024-01-05	medium	closed	definition agreed
Quillon Home Supply - second site	aisle congestion at peak	2023-12-27	high, customer executive involved	monitoring	parameter set applied
Tenmile Freight - second site	charger fault on two units	2023-12-18	medium	escalated to product	parameter set applied
Bexley Pharmaceutical Wholesale - second site	interface timeouts to the host	2023-12-09	high, customer executive involved	open	definition agreed

Spares, depots and warranty

Engineers are quietly building their own vehicle stock, which means the depot numbers describe a system nobody uses. Vehicle stock is masking the true fill rate. Fill rate ran 90 percent against 85 percent. Here the queue was read account by account. Only one hub held tooling then. Depot fill rate was 90 percent. Ingrid Solberg approved the critical spares increase. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Depot placement was reconsidered, because half of the courier spend goes to sites within a day's drive of a hub. Depot fill rate was 90 percent. Return material flow remains slow. Here the queue was read account by account. Only one hub held tooling then. Salinas Valley took the planned sensor head replacements. Vehicle stock is masking the true fill rate. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

The recall-style replacement of affected sensor heads is proceeding on a planned schedule rather than on failure. Vehicle stock is masking the true fill rate. Depot fill rate was 90 percent. Here the queue was read account by account. Only one hub held tooling then. Salinas Valley took the planned sensor head replacements. Return material flow remains slow. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Service parts pricing has never been reviewed, and two parts are being sold below their landed cost. Fill rate ran 90 percent against 85 percent. Salinas Valley took the planned sensor head replacements. Here the queue was read account by account. Only one hub held tooling then. Return material flow remains slow. Depot fill rate was 90 percent. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Obsolescence on the first-generation controller will force a spares buy decision before the end of next year. Depot fill rate was 90 percent. Salinas Valley took the planned sensor head replacements. Here the queue was read account by account. Only one hub held tooling then. Warranty cost per fleet unit is falling. Ingrid Solberg approved the critical spares increase. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

First-generation controllers are heading for obsolescence, and the spares buy has to be decided before the line stops making them. Vehicle stock is masking the true fill rate. Depot fill rate was 90 percent. Here the queue was read account by account. Only one hub held tooling then. Return material flow remains slow. Ingrid Solberg approved the critical spares increase. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Constraint tracking since the last session

Constraint under review: field engineering capacity, at the gap identified stage as of this session. Deployment throughput is not limited by robot supply this quarter; it is limited by the number of field engineers certified to sign off a site acceptance test without a supervising lead. Work in process 31 units, mostly Lidar assemblies. The HS-400 pallet mover took most of the test cell time.

Expedite spend is the honest measure of how well planning is working, and it went the wrong way this period. Brightwater Motion improved after a site visit. Coverage weeks on long-lead parts were extended. Here the queue was read account by account. Only one hub held tooling then. Penrose Castings missed two consecutive commit dates. On-time delivery was 88 percent, from 86 percent. Go-live cycle 59 days at Salinas Valley. Queue depth 11 sites, Cedar Bluff at the front. Contract crew Verrado closed Grand Forks without a second visit. Requisitions open 19, Tobias Nkemdirim holding the list.

One casting supplier has become a constraint on the pallet mover, and the tooling is theirs rather than ours. Brightwater Motion improved after a site visit. Expedite activity rose with the build plan of 48. Here the queue was read account by account. Only one hub held tooling then. Supplier on-time delivery came in at 88 percent. On-time delivery was 88 percent, from 86 percent. The HS-400 pallet mover took most of the test cell time. Ingrid Solberg and Nadia

Oyelaran disagreed on sequencing. Uptime 96.1 percent, interventions every 281 hours. Attrition 10 percent annualised in Southeast.

Spares depot performance by part class, Q1 2024

Depot	Part class	Part lines stocked	Fill rate %	Backorders	Repair backlog units
Reno depot	mechanical	134	94	7	21
Columbus depot	mechanical	213	89	0	0
Dallas depot	mechanical	166	96	8	15
Atlanta depot	mechanical	189	85	3	5
Toronto depot	mechanical	212	93	10	19
Vehicle stock	mechanical	179	88	5	1
Reno depot	electrical and sensing	258	95	12	11
Columbus depot	electrical and sensing	253	86	1	0
Dallas depot	electrical and sensing	234	91	9	13
Atlanta depot	electrical and sensing	229	84	4	3
Toronto depot	electrical and sensing	280	92	11	17
Vehicle stock	electrical and sensing	275	87	2	7

Headcount and recruiting

The recruiting funnel converts poorly at the on-site stage for field roles, and the loop design is the likely reason. Annualised attrition sits at 10 percent. Open requisitions number 19. Here the queue was read account by account. Only one hub held tooling then. Median time to fill was 82 days. Headcount stood at 125 against a plan of 131. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Internal mobility filled two roles this period, which is cheaper and faster than external hiring and creates a backfill. The headcount gap is 6 roles. Requisitions fell from 17 to 19. Here the queue was read account by account. Only one hub held tooling then. Open requisitions number 19. Time to fill improved from 76 to 82 days. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Performance calibration ran across all functions for the first time, and the distribution differed sharply between engineering and operations. Annualised attrition sits at 10 percent. Headcount stood at 125 against a plan of 131. Here the queue was read account by account. Only one hub held tooling then. The headcount gap is 6 roles. Median time to fill was 82 days. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Recruiting moved to a regional pipeline model with standing phone screens, which changed time to fill more than any sourcing spend did. Headcount stood at 125 against a plan of 131. Requisitions fell from 17 to 19. Here the queue was read account by account. Only one hub held tooling then. Open requisitions number 19. Median time to fill was 82 days. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Attrition is low in aggregate and uncomfortable in specifics, because the leavers include two people the company cannot easily replace. Annualised attrition sits at 10 percent. Open requisitions number 19. Here the queue was read account by account. Only one hub held tooling then. Median time to fill was 82 days. The headcount gap is 6 roles. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

Compensation bands for field engineering were refreshed after three offers were declined on base salary. Annualised attrition sits at 10 percent. Requisitions fell from 17 to 19. Here the queue was read account by account. Only one hub held tooling then. Median time to fill was 82 days. Open requisitions number 19. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Cash conversion

Deposit terms on new multi-site commitments materially change the working capital profile of the site queue. Collections totalled 6110 thousand dollars. Days sales outstanding closed at 58 days. Here the queue was read account by account. Only one hub held tooling then. Outstanding days improved from 55 to 58. Nadia Oyelaran moved collections calls to the account teams. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Customer-side accounts payable portals are a real source of delay, and three accounts require a different submission path. Nadia Oyelaran moved collections calls to the account teams. Two receivables aged past ninety days. Here the queue was read account by account. Only one hub held tooling then. Alder & Vance Retail requires a separate submission path. Collections totalled 6110 thousand dollars. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Progress billing on long deployments would smooth the cash profile, and the commercial team has resisted it for good reasons. Two receivables aged past ninety days. Alder & Vance Retail requires a separate submission path. Here the queue was read account by account. Only one hub held tooling then. Receipts of 6110 thousand outpaced invoicing. Collections totalled 6110 thousand dollars. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Aged balances were read out account by account, which took ten minutes and closed two of them the same week. Days sales outstanding closed at 58 days. Two receivables aged past ninety days. Here the queue was read account by account. Only one hub held tooling then. Alder & Vance Retail requires a separate submission path. Nadia Oyelaran moved collections calls to the account teams. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Days sales outstanding is the metric under active management, and the improvement came from the invoicing process rather than from customer behaviour. Receipts of 6110 thousand outpaced invoicing. Outstanding days improved from 55 to 58. Here the queue was read account by account. Only one hub held tooling then. Collections totalled 6110 thousand dollars. Two receivables aged past ninety days. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

Two large receivables aged past ninety days, and both trace to a purchase order number mismatch rather than to unwillingness to pay. Days sales outstanding closed at 58 days. Receipts of 6110 thousand outpaced invoicing. Here the queue was read account by account. Only one hub held tooling then. Two receivables aged past ninety days. Outstanding days improved from 55 to 58. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Warranty claims by build vintage and line, Q1 2024

Build vintage	Product line	Fleet units	Claims in period	Warranty cost per unit USD	Trend
2023 build	HS-200 tote handler	58	10	312	improving sharply
2024 first half build	HS-200 tote handler	72	7	290	first period measured
2024 second half build	HS-200 tote handler	101	17	246	improving
2025 first half build	HS-200 tote handler	130	9	224	flat
2025 second half build	HS-200 tote handler	147	19	290	worsening
2026 build	HS-200 tote handler	188	18	268	improving sharply
2023 build	HS-400 pallet mover	55	11	202	first period measured
2024 first half build	HS-400 pallet mover	75	4	312	improving
2024 second half build	HS-400 pallet mover	98	11	224	flat
2025 first half build	HS-400 pallet mover	133	8	246	worsening
2025 second half build	HS-400 pallet mover	156	19	268	improving sharply

Build vintage	Product line	Fleet units	Claims in period	Warranty cost per unit USD	Trend
2026 build	HS-400 pallet mover	173	16	202	first period measured

Runway and triggers

Contract labour in deployment is the fastest lever available and the one with the most direct customer consequence. Runway stands at 24 months on the middle case. Test-cell capital was deferred again. Here the queue was read account by account. Only one hub held tooling then. Burn for the period was 7290 thousand dollars. Runway moved from 25 to 24 months. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Cash discipline this year came two-thirds from collections and one-third from deferred capital, which is a better mix than cutting people. Test-cell capital was deferred again. Priya Ramanathan holds the trigger list. Here the queue was read account by account. Only one hub held tooling then. Capital spend was 1120 thousand dollars. Burn for the period was 7290 thousand dollars. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Runway is not a single number, and presenting it as one has caused more confusion than it has prevented. Runway stands at 24 months on the middle case. Runway moved from 25 to 24 months. Here the queue was read account by account. Only one hub held tooling then. Burn for the period was 7290 thousand dollars. Spending ran 7290 thousand against 2455 thousand. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Each scenario was read against a named observable, so nobody has to interpret whether a trigger has fired. Test-cell capital was deferred again. Capital spend was 1120 thousand dollars. Here the queue was read account by account. Only one hub held tooling then. Spending ran 7290 thousand against 2455 thousand. Burn for the period was 7290 thousand dollars. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Capital spending on test cells was deferred, which protects cash now and constrains build capacity later. Runway stands at 24 months on the middle case. Burn for the period was 7290 thousand dollars. Here the queue was read account by account. Only one hub held tooling then. Capital spend was 1120 thousand dollars. Runway moved from 25 to 24 months. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

The downside trigger was armed after two large bookings left the quarter, and nine requisitions were paused within a week. Spending ran 7290 thousand against 2455 thousand. Runway moved from 25 to 24 months. Here the queue was read account by account. Only one hub held tooling then. Capital spend was 1120 thousand dollars. Priya Ramanathan holds the trigger list. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Headcount plan against actual by function and location, Q1 2024

Function	Location	Plan	Actual	Gap	Open requisitions
Engineering	headquarters and plant	8	7	2	6
Manufacturing	headquarters and plant	3	3	0	1
Deployment and field service	headquarters and plant	3	3	0	0
Commercial	headquarters and plant	1	1	0	0
Customer success	headquarters and plant	24	23	3	3
Supply chain	headquarters and plant	17	17	1	3
Finance and administration	headquarters and plant	14	13	2	4
People	headquarters and plant	9	10	0	2

Function	Location	Plan	Actual	Gap	Open requisitions
Engineering	field and regional	4	3	1	5
Manufacturing	field and regional	2	2	0	0
Deployment and field service	field and regional	1	1	0	0
Commercial	field and regional	1	1	0	0
Customer success	field and regional	14	12	1	2
Supply chain	field and regional	10	9	0	2
Finance and administration	field and regional	8	7	1	3
People	field and regional	5	5	0	1

Product and roadmap checkpoints

Product management brought a decision to stop building one feature, and the review supported stopping it. Tobias Nkemdirim brought the decision to stop one feature. Field data now sets the reliability backlog order. Here the queue was read account by account. Only one hub held tooling then. Adapter work crowded out platform work again. Two roadmap checkpoints cleared; one slipped. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Simulation coverage before release improved, and the last two releases produced fewer field regressions as a result. The HS-400 pallet mover absorbed most of the engineering effort. Field data now sets the reliability backlog order. Here the queue was read account by account. Only one hub held tooling then. Documentation lags the current release. Tobias Nkemdirim brought the decision to stop one feature. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Stopping work is harder than starting it, and this session stopped one thing on purpose. Adapter work crowded out platform work again. Tobias Nkemdirim brought the decision to stop one feature. Here the queue was read account by account. Only one hub held tooling then. Release staging across sites is now standard. Field data now sets the reliability backlog order. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Roadmap checkpoints were reviewed against the commitments made to customers rather than against the internal plan, which is a harder test. The HS-400 pallet mover absorbed most of the engineering effort. Field data now sets the reliability backlog order. Here the queue was read account by account. Only one hub held tooling then. Two roadmap checkpoints cleared; one slipped. Documentation lags the current release. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Hardware and software roadmaps are coupled more tightly than the planning process admits, and one dependency was found late. Tobias Nkemdirim brought the decision to stop one feature. Adapter work crowded out platform work again. Here the queue was read account by account. Only one hub held tooling then. The HS-400 pallet mover absorbed most of the engineering effort. Release staging across sites is now standard. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

The next controller revision closes an obsolescence exposure and requires a re-validation cycle that has to be planned now. Field data now sets the reliability backlog order. Tobias Nkemdirim brought the decision to stop one feature. Here the queue was read account by account. Only one hub held tooling then. Release staging across sites is now standard. The HS-400 pallet mover absorbed most of the engineering effort. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Where the room did not agree

The plant manager objected to another plan revision inside the month and asked for a freeze window, which was granted in part. The minority view is preserved in the record. Priya Ramanathan and Tobias Nkemdirim read the data

differently. Here the queue was read account by account. Only one hub held tooling then. A further test was added because of the dissent. No consensus was reached; the Chief Executive Officer decided. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Whether a deferral saves money depends on which function is counting, and both counts were tabled. A further test was added because of the dissent. The minority view is preserved in the record. Here the queue was read account by account. Only one hub held tooling then. The disagreement is recorded rather than resolved. The objection was noted against the decision. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Disagreement was recorded rather than smoothed over, because the two positions rest on different readings of the same data. The objection was noted against the decision. No consensus was reached; the Chief Executive Officer decided. Here the queue was read account by account. Only one hub held tooling then. Priya Ramanathan and Tobias Nkemdirim read the data differently. A further test was added because of the dissent. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Finance and deployment disagreed about whether contract labour should be treated as a fixed or variable commitment. Priya Ramanathan and Tobias Nkemdirim read the data differently. A further test was added because of the dissent. Here the queue was read account by account. Only one hub held tooling then. The disagreement is recorded rather than resolved. The minority view is preserved in the record. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Two functions read the reliability data differently, and the review agreed to a single definition before arguing again. The minority view is preserved in the record. Priya Ramanathan and Tobias Nkemdirim read the data differently. Here the queue was read account by account. Only one hub held tooling then. A further test was added because of the dissent. No consensus was reached; the Chief Executive Officer decided. Depot fill 90 percent at Dallas depot. Scrap 26 thousand, Brightwater Motion contributing. Ingrid Solberg and Nadia Oyelaran disagreed on sequencing. Work in process 31 units, mostly Lidar assemblies.

Product and sales disagreed on whether the pilot feature should be committed to a customer date, and the answer was no. Nadia Oyelaran asked for the decision to be revisited. The objection was noted against the decision. Here the queue was read account by account. Only one hub held tooling then. No consensus was reached; the Chief Executive Officer decided. The disagreement is recorded rather than resolved. Queue depth 11 sites, Cedar Bluff at the front. Attrition 10 percent annualised in Southeast. The HS-400 pallet mover took most of the test cell time. Headcount 125 of 131, Priya Ramanathan reporting.

Requisition ageing and decisions, Q1 2024

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
REQ-0340	Field engineer	90	offer out	hiring manager sourced	continue
REQ-0342	Crew lead	71	offer declined	external agency	raise band
REQ-0344	Controls engineer	100	on hold	external agency	widen radius
REQ-0346	Manufacturing technician	81	closed without hire	external agency	pause
REQ-0348	Buyer and planner	110	sourcing	internal recruiting	close and re-scope
REQ-0350	Solutions architect	67	phone screen	external agency	convert contractor
REQ-0352	Service dispatcher	94	on-site loop	hiring manager sourced	continue
REQ-0354	Reliability engineer	75	offer out	internal recruiting	raise band
REQ-0356	Field engineer (second opening)	104	offer declined	external agency	widen radius
REQ-0358	Crew lead (second opening)	69	on hold	Renata Voss	pause
REQ-0360	Controls engineer (second opening)	90	closed without hire	external agency	close and re-scope
REQ-0362	Manufacturing technician (second	63	sourcing	external agency	convert contractor

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
	opening)				
REQ-0364	Buyer and planner (second opening)	90	phone screen	internal recruiting	continue
REQ-0366	Solutions architect (second opening)	71	on-site loop	internal recruiting	raise band
REQ-0368	Service dispatcher (second opening)	100	offer out	internal recruiting	widen radius
REQ-0370	Reliability engineer (second opening)	81	offer declined	internal recruiting	pause
REQ-0372	Field engineer (backfill)	110	on hold	internal recruiting	close and re-scope

Decisions, owners and dates recorded in Q1 2024

Items closed this period were closed with named evidence: a report, a signed document or a measured result. Renata Voss, VP People, took the largest open item. The oldest item dates from two sessions back. Here the queue was read account by account. Only one hub held tooling then. One item closed with measured evidence. Two due dates were changed in the room. Bookings 142 units across 86 pursuits. Yield 93 percent, scrap 26 thousand in Frame and drive assembly. Headcount 125 of 131, Priya Ramanathan reporting. Uptime 96.1 percent, interventions every 281 hours.

Cross-functional items were given a single owner rather than two, after a period where shared ownership meant no ownership. Owners were named and dates set against 2024-05-16. One item closed with measured evidence. Here the queue was read account by account. Only one hub held tooling then. Three items carried forward into 2024-05-16. Two due dates were changed in the room. Dallas depot cleared its repair backlog. Scrap 26 thousand, Brightwater Motion contributing. Slipped deals 9, including Alder & Vance Retail. On-time delivery 88 percent, Kestrel Drive Systems the outlier.

Evidence of closure means a document, a measurement or a signature, and status colours no longer qualify. Two due dates were changed in the room. Three items carried forward into 2024-05-16. Here the queue was read account by account. Only one hub held tooling then. Owners were named and dates set against 2024-05-16. One item closed with measured evidence. Pinebank Distribution asked for the Beacon WMS adapter configuration. Uptime 96.1 percent, interventions every 281 hours. Capital spend 1120 thousand this period. Norfolk carries 11 of the queued fleet.

Dates were negotiated with the owners present, which removed the usual round of silent renegotiation afterwards. Ingrid Solberg inherited an item after a role change. Two due dates were changed in the room. Here the queue was read account by account. Only one hub held tooling then. The oldest item dates from two sessions back. Renata Voss, VP People, took the largest open item. On-time delivery 88 percent, Kestrel Drive Systems the outlier. Go-live cycle 59 days at Salinas Valley. The HS-400 pallet mover took most of the test cell time. Pinebank Distribution asked for the Beacon WMS adapter configuration.

Open and closed actions carried through Q1 2024

Reference	Item	Owner	Original due	Current due	Disposition
OR-0307	Raise geographic radius on field engineering requisitions	Renata Voss	2024-04-25	2024-05-16	Open; none yet
OR-0320	Reconcile the two uptime definitions with the customer	Owen Castellanos	2024-04-15	2024-05-06	Blocked; outstanding
OR-0323	Refresh compensation bands for field engineering	Renata Voss	2024-04-21	2024-05-12	Closed by decision; decision recorded
OR-0326	Move collections calls to the account teams	Lucia Mbeki-Hart	2024-04-27	2024-05-18	Open, date revised; outstanding
OR-0329	Split the reliability backlog by build vintage	Tobias Nkemdirim	2024-05-03	2024-05-24	Blocked; outstanding
OR-0332	Cost the second-shift ramp against overtime	Marcus Peil	2024-05-09	2024-05-30	Closed by decision; report filed
OR-0335	Reissue the deployment playbook	Priya Ramanathan	2024-05-15	2024-06-05	Open, date revised; outstanding

Reference	Item	Owner	Original due	Current due	Disposition
	network section				
OR-0338	Draft deposit terms for multi-site commitments	Ingrid Solberg	2024-05-21	2024-06-11	Blocked; outstanding
OR-0341	Stage software releases across sites rather than at once	Griffin Tasse	2024-05-27	2024-06-17	Closed by decision; report filed
OR-0344	Rebuild the site cost model separating retrofit from greenfield	Marcus Peil	2024-06-02	2024-06-23	Open, date revised; outstanding
OR-0347	Measure onboarding time to productivity for crews	Renata Voss	2024-06-08	2024-06-29	Blocked; outstanding
OR-0350	Complete process capability study on the bond operation	Emil Farkas	2024-06-14	2024-07-05	Closed by decision; decision recorded
OR-0353	Size the buffer stock on long-lead commodities	Hollis Bramwell	2024-06-20	2024-07-11	Open, date revised; outstanding
OR-0356	Read the tooling ownership clauses on four part numbers	Hollis Bramwell	2024-06-26	2024-07-17	Blocked; outstanding
OR-0359	Set a plan freeze window inside the build month	Emil Farkas	2024-07-02	2024-07-23	Closed by decision; decision recorded